Homework 3

Due: 2/12

- You must show all calculations and important intermediate results. Otherwise, you will lose points even if your answers are correct.
- If you use R, you must submit the R code file.

Problem 1 (10 points). This problem is about the smoothing by binning, which we discussed in the class. Consider the following variable:

<16, 22, 33, 42, 49, 57, 58, 66, 79, 83, 88, 95, 113, 128>

(1). Smooth the variable using the equal-width binning method and bin means. Use three bins.
(2). Smooth the variable using the equal-width binning method and bin boundaries. Use three bins.

Problem 2 (10 points). Use the *ThoracicSurgery.csv* file for this problem. Consider the following five attributes: *age*, *trestpbs*, *chol*, *thalch*, *oldpeak*.

Create a correlation matrix that shows correlation between all pairs of attributes. Which pair of attributes has the highest correlation?

You may use any tool, including R, for this problem.

Problem 3 (10 points). Use the *ThoracicSurgery.csv* file for this problem. Determine whether there is a correlation between *cp* and *diagnosis* using the chi-square test method that we discussed in the class. You may use any tool to derive a contingency table. However, you must do all calculations yourself after that, including the calculation of expected values and the test statistic. You may use a spreadsheet software or R just for the purpose of calculation.

Problem 4 (10 points). Use the *hw3_p4.csv* file. Use R for this problem.

(1). Select a subset of attributes using the *cfs* method and show the selected attributes.
(2). Select a subset of attributes using the *Boruta* method and show the selected attributes.
(3). Select the top 5 attributes using the *information gain* method and show the selected attributes.

Include all answers in a single Word or PDF document and upload it to Blackboard. Use *LastName_FirstName_hw*3*.docx* or *LastName_FirstName_hw*3*.pdf* as the file name. If you have additional files, such as an Excel file or a R code file, then combine all of them into a single archive file and name it *LastName_FirstName_hw*3*.EXT*, where *EXT* is an appropriate archive file extension such as *zip* or *rar*.